Defend — Full Review

FORMULA

Thymosin Alpha-1 5mg + KPV 2mg + LL-37 1mg

What's in it

PEPTIDE	DOSE	ROLE IN THIS BLEND
Thymosin Alpha-1	5mg	The coordinator. Recruits and regulates T-cells for a faster, more organized immune response, rather than just a stronger one.
KPV	2mg	The calibrator. Turns down inflammation that's run too high, without blunting the immune system's normal job.
LL-37	1mg	The direct responder. A broad-spectrum antimicrobial peptide that acts on pathogens itself, rather than only signaling other cells to respond.

The combination logic

The three work at different points in how the immune system responds. Thymosin Alpha-1 helps coordinate the immune cells that do the actual defending. KPV turns down inflammation that's run too high, without shutting down the immune system's normal job. LL-37 acts directly on pathogens themselves. Together, the idea is coordination, calibration, and direct defense.

Clinical detail: Thymosin Alpha-1 is a biological response modifier that activates T-lymphocytes and natural killer cells and raises interleukin-2 and interferon-gamma, while also calming pro-inflammatory cytokine excess — a dual immune-modulating action. KPV (the C-terminal tripeptide of alpha-MSH) inhibits NF-κB activation, the transcription factor driving inflammatory cytokine production, without broadly suppressing immune competence. LL-37 is a cathelicidin-derived peptide with a cationic, amphipathic structure that disrupts microbial membranes directly, while also recruiting immune cells and promoting wound re-epithelialization. The three are proposed to act on non-competing points: coordination, inflammatory calibration, and direct antimicrobial action.

Is there evidence for the combination, or only for the parts?

> Thymosin Alpha-1 has the most developed evidence of the three — it is approved in some countries outside the US for immune-related indications, though not FDA-approved domestically. KPV and LL-37 both have supportive animal and mechanistic data but limited human trial evidence for the uses commonly marketed. No study has tested this specific three-peptide combination; the rationale for combining them is mechanistic layering, not comparative trial evidence.

What it's used for

USED FOR	EVIDENCE	IN SHORT
General immune support	human_limited / animal_only	Thymosin Alpha-1 has the most developed evidence of the three, including approved uses outside the US for immune modulation.

USED FOR	EVIDENCE	IN SHORT
Inflammation control without immune suppression	animal_only	KPV's proposed mechanism specifically avoids blunting normal immune defense while calming excess inflammatory signaling.
Antimicrobial / wound-related support	animal_only	LL-37's primary studied role; broad-spectrum activity against bacteria, viruses, and fungi in preclinical research.

Overlaps & stacked risks

- Combining three active immune-modulating peptides carries the summed risk profile of all three.
- LL-37 can cause injection-site irritation or mild inflammation; caution in patients with autoimmune conditions or high baseline inflammation.
- Long-term continuous use of LL-37 is specifically not recommended without provider oversight.
- No long-term human safety data exists for this combination.

Who it's not for

- Pregnant or breastfeeding patients
- Patients with autoimmune disorders or high baseline inflammation, without clinician oversight
- Anyone with a highly reactive immune history — monitor for unexpected sensitivity
- Long-term continuous use without provider oversight is not recommended for LL-37 specifically

Who it suits

Patients seeking general immune support — during high-exposure periods, recovery from illness, or persistent low-grade infection concerns — who have no autoimmune history and have discussed the limited evidence base with their provider.

Who makes this, and under what rules

Everything dispensed here is prepared by a licensed 503A compounding pharmacy — not bought from a supplier and repackaged, and not sourced from an online seller.

That designation carries specific, enforceable obligations:

- **A prescription in your name.** Compounded from a prescription for one identified patient, after an evaluation.
- **Licensed and inspected.** A state licence, subject to inspection by the state board of pharmacy.
- **Ingredients that can be traced.** A certificate of analysis from an FDA-registered facility, back to a named source and lot.
- **Sterile preparation to a published standard.** USP standards.
- **A paper trail.** A lot number and beyond-use date on every preparation.

A compounded preparation is not an FDA-approved medicine — but it is the difference between a licensed pharmacy working to a standard and an unregulated website.

This page describes Defend for general educational purposes. It is not medical advice and does not replace a consultation with a licensed provider. Individual results vary, and combination products carry the risks of each included peptide.